

Generative AI, Chatbots, and Programming Skills: Examining Code Suggestion Features in Enhancing Self-learning Capabilities

Abstract

This study investigates the role of generative artificial intelligence (AI) chatbots, specifically their code suggestion features, in enhancing the self-learning capabilities of university students in programming. With the backdrop of a declining student population in developed East Asian countries and a corresponding increase in international students, this research examines how AI can revolutionize programming education in universities, focusing on big data analytics. The study assesses the effectiveness of AI tools in facilitating personalized learning experiences. Using a mixed-methods approach involving quantitative analysis and interviews, the study compares student performance in programming tasks with and without AI assistance. The results indicate a significant improvement in academic performance among students who used AI chatbots for learning, highlighting the potential of AI in enhancing understanding and practical application of programming concepts. Furthermore, the study explores the ethical implications and the need for an interactive, student-centered approach in curriculum design. The findings suggest that while generative AI chatbots can significantly aid in programming education, they should be integrated thoughtfully, considering the challenges of AI-generated content and the importance of human oversight in educational settings.

Keywords: generative AI, programming education, self-learning, AI chatbots

1. Introduction

The integration of Generative artificial intelligence (AI), particularly ChatGPT by OpenAI, has emerged as a pivotal development in various domains, including higher education. As highlighted by Dowling and Lucey (Dowling & Lucey, 2023), Eke (Eke, 2023), and Lim et al. (Lim, Gunasekara, Pallant, Pallant, & Pechenkina, 2023), these AI systems, through their data-driven content generation capabilities, are influencing diverse areas. ChatGPT, specifically, leverages deep learning to produce natural language text, trained on a wide array of online text data. This ability to generate human-like language text positions it as a valuable tool for numerous writing tasks.

The International Journal of Management Education (IJME) recently highlighted the impact of new Generative AI on improving educational and business training opportunities. This discussion emphasizes AI's role in transforming learning methods and enhancing the efficacy of business education. Such advancements align with the journal's focus on integrating emerging technologies in management training paradigms. For example, Lim et al. (Lim et al., 2023) delve into the role of Generative AI in the field of education and its potential impacts on the future of learning. Their study underscores the dual nature of Generative AI as both a catalyst for educational reform and a source of significant challenges. This research by Lim et al. (Lim et al., 2023) represents a crucial contribution in understanding the transformative implications of Generative AI in education from the perspective of management educators. Duong et al.

(Duong, Vu, & Ngo, 2023) explore higher education students' use of ChatGPT through a modified technology acceptance model with highlighting the significant roles of effort expectancy and knowledge sharing in influencing both the intention and actual use of ChatGPT. This research provides vital insights for understanding and integrating Generative AI in educational contexts.

The ongoing decrease in the proportion of university students in developed East Asian countries, such as South Korea, marks a transformative shift in the global higher education landscape with a growing emphasis on sustainability (Hyun Yoo, 2023). Korean universities exemplify the novel obstacles and prospects that higher education institutions are confronting in their pursuit of sustainable educational models and practices (J. Kim & Ilon, 2020). This shift reflects a broader recognition of the need for sustainable approaches in adapting to changing demographics and educational demands (Y. Wang & Fukuda, 2019). To ensure their survival, it is imperative for higher education institutions to move away from the traditional approach of focusing on quantity and instead prioritize cultivating a diverse and dynamic learning environment (Goralski & Tan, 2020).

Furthermore, the declining student population in South Korea serves as a precursor to an anticipated rise in international students entering the country (Koh, 2023). This transition could have significant implications not only for South Korea but also for other nations grappling with aging populations, including Japan, the United Kingdom, and France. Consequently, there is mounting interest in exploring whether leveraging modern information technology and artificial intelligence-based applications within programming education can serve as a viable technological alternative to address the current challenges faced by Korean universities. Specifically, there is a need to assess whether these advancements can effectively train professionals specializing in big data and machine learning (Hong, 2017).

The utilization of generative artificial intelligence applications like ChatGPT and AI-based translation services, such as DeepL and Google Translator, is already evident in the field of education (Daun & Brings, 2023; Loos, Gröpler, & Goudeau, 2023; Rawas, 2023; Zhu, Sun, Luo, Li, & Wang, 2023). The increasing interest in the potential of generative AI in education and related sectors can be attributed to the shifting usage patterns among not only international- but also domestic students (T. Wang et al., 2023). Rather than relying solely on AI applications for assignment completion or class attendance, these students are now turning to platforms such as ChatGPT's code assistant features and GitHub's Copilot for information retrieval or programming assistance. However, there remains a dearth of comprehension regarding how generative AI can enhance educational administration, curriculum development, pedagogy, and student learning specifically within the realm of data science education.

This study aims to address the research gap surrounding the impact of artificial intelligence on machine learning and big data programming education in universities, as well as explore how it can be utilized to enhance teaching methods. By investigating these research questions, we can gain insights into the potential benefits of artificial intelligence in facilitating personalized learning experiences for students. Moreover, this study recognizes the importance of instructors adopting a more interactive and student-centered approach to curriculum design, moving away from traditional one-way lectures towards a more organic and engaging learning environment. Additionally, this research also acknowledges the need to develop strategies for effectively interpreting and responding to AI-generated

suggestions, explanations, and modifications of student-written code, as well as addressing challenges related to hallucination problems exhibited by generative AI. The findings of this study will contribute towards establishing specific guidelines that can inform educational practices and help overcome these individual challenges within the field.

2. Background

2.1. Higher Education and Challenges in South Korea

South Korea has experienced rapid economic growth in recent decades. Following the devastation of the Korean War in the 1950s, the country initially relied heavily on agricultural, forestry, and mining sectors due to a lack of industrial infrastructure. However, South Korea swiftly embarked on a path of industrialization and achieved remarkable progress. Notably, it made significant advancements in telecommunications, internet services, and entertainment industries. This development propelled South Korea into the ranks of developed nations by 2020.

Meanwhile, the most significant challenges that could hinder South Korea's long-term success are its declining and aging population. South Korea heavily relies on imported raw materials for its economy, particularly in processing these materials to make products. This necessitates the utilization of advanced technologies to enhance their value, thus highlighting the importance of a consistent supply of highly educated workers in the industry. However, South Korea is currently grappling with a rapidly decreasing population. The effectiveness of educational methods employed during the baby boom era is diminishing, resulting in a decline in the quality of education services. Given the increasing significance of higher education, enhancing individual student performance has become an imperative matter.

According to a 2023 analysis conducted by the South Korean Ministry of Education and the Korea Education Development Institute, not only has the number of higher education institutions teaching students decreased, but the number of students enrolled in education has been on the decline since 2011 (KOSIS, 2023). Since 2021, the number of international students has increased for three consecutive years and now exceeds 180,000 (Son, 2023). The number of students enrolled in higher education institutions decreased by 17.1% from 2014 to 2023. The number of international students rose from approximately 50,000 in 2014 to around 130,000 in 2023. During this time, tuition fees at South Korean universities were suspended for political reasons, so the influx of international students became the primary means by which universities overcame financial difficulties. This situation is anticipated to deteriorate.

According to the latest statistics from South Korea's Ministry of Trade, Industry and Energy (MOTIE) and the Korea Institute of Advanced Industrial Technology (KIAT), the ratio of industrial technology workers to the total number of workers was 34.1% in 2018, up 1.7% from the year before the survey (KIAT, 2019). Among the industries driving new business growth in Korea, ICT-related sectors such as software and electronics have driven this increase, despite a decreasing labor force, with a 72.3% shortage in 2015 and a 73.3% shortage in 2017. The software industry contributed 21.8% and the electronics industry contributed 18.8%. Because of Korea's increasing reliance on the ICT industry, it is unable to provide companies with sufficient workers with advanced degrees. This, coupled with the worsening financial situation of universities and the deteriorating quality of higher education, is a challenge that will be difficult to surmount soon.

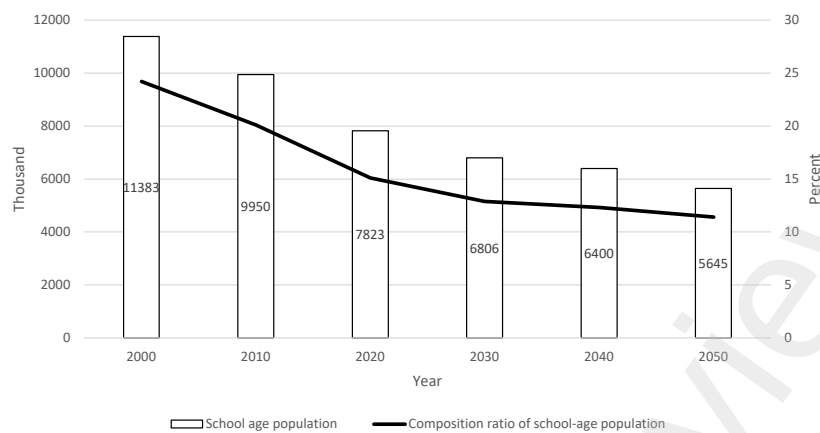


Figure 1. Changes in South Korea's school-age population (StatisticsKorea, 2023)

2.1. Current Applications of AI in Education

The use of AI in the area of education is on the rise, with the potential to enhance learning experiences and personalize educational content to individual students. As a result, there has been a notable increase in research endeavors aimed at leveraging AI for the improvement of education. According to the findings of Seldon and Abidoye (Seldon & Abidoye, 2018), artificial intelligence (AI) has proven to be effective in enhancing learning experiences and streamlining the delivery and customization of educational materials to meet individual student requirements. The aforementioned phenomenon has prompted a notable increase in scholarly investigations (e.g., (Chichekian & Benteux, 2022)) focused on leveraging AI to improve educational methodologies. Through the utilization of various computing technologies, including intelligent education, innovative virtual learning, and data analysis and prediction, artificial intelligence (AI) has the potential to improve the quality and effectiveness of educational processes (L. Chen, Chen, & Lin, 2020; Kahraman, Sagiroglu, & Colak, 2010).

AI has emerged as a promising tool for enhancing the value and performance of learning (X. Chen, Xie, Zou, & Hwang, 2020). By integrating information technologies such as intelligent learning planners, virtual instructors, and data analysis and prediction algorithms, AI has the potential to revolutionize education. Various AI-assisted educational services, including ActiveMath, MATHia, and Why2Atlas, have been developed as interactive learning environments that not only monitor student progress but also facilitate goal management, enhance teaching tools, and promote global communication and feedback between instructors and students (Chassignol, Khoroshavin, Klimova, & Bilyatdinova, 2018). Additionally, AI-driven applications like Grammarly, Ecree, and PaperRater have proven beneficial for foreign students by assisting them with administrative task (Johnson, Nádas, & Bell, 2010).

Research has demonstrated the efficacy of learner-centered AI applications such as Deep Tutor and Auto Tutor in tailoring content to the specific abilities and needs of university students (Rus, D'Mello, Hu, & Graesser, 2013). By providing customized and personalized learning experiences, these AI tools have the potential to enhance educational outcomes and potentially contribute to improved academic performance (Mikropoulos & Natsis, 2011; Sharma, Kawachi, & Bozkurt, 2019). Moreover, the advent of online learning has facilitated the development of AI-powered

teaching tools that can transcend geographical boundaries, enabling international students to benefit from language translation technologies like Google Translate. A study conducted by Tsai (Tsai, 2019) on Chinese college students learning English revealed that Google Translate assisted foreign learners in utilizing more advanced vocabulary and reducing spelling and grammar errors. Additionally, employing virtual reality, 3D technologies, and gamification techniques in instruction has not only elevated teaching standards but also increased efficiency in catering to the needs of international students (Lee & Hammer, 2011).

Numerous scholarly investigations have substantiated the capacity of AI to augment the educational experiences of students. Chassignol et al. (Chassignol et al., 2018) have conducted a study on the utilization of AI in various educational applications, including Knewton, Cerego, Immersive Reader, and CALL. These applications employ machine learning algorithms to offer immediate recommendations and customize educational resources to cater to the specific requirements of learners across different educational stages, spanning from preschool to graduate school. In a similar vein, Pokrivcáková (Pokrivcakova, 2019) has emphasized the efficacy of chatbots that leverage AI algorithms to tailor learning materials to the specific needs and capabilities of students. Furthermore, the utilization of AI technologies such as Pearson's Write-to-Learn and Turnitin has been implemented to enhance academic integrity through the provision of support for editing and writing activities, in addition to supporting tailored learning experiences. Nevertheless, Crowe et al. (Crowe, LaPierre, & Kebritchi, 2017) express concern regarding the potential unintended consequences of these tools, as they may unwittingly promote unethical behaviors by guiding students towards websites or platforms that facilitate the process of creating papers, so potentially undermining the principles of academic integrity.

The integration of AI powered translators and writing tools, voice recognition and dictation tools, as well as language learning tools, can potentially enhance the language acquisition journey for non-native English-speaking international students, especially when pursuing education in English-speaking countries. An illustrative instance is the combination of Google Translate, Grammarly and even ChatGPT, an application designed to aid in writing tasks, which can facilitate the process of composing academic content for international students, faculty members, and researchers within an academic environment (ONeill & Russell, 2019; Pratama, 2021). Additionally, other resources like ProWritingAid, Quillbot, and Ginger offer valuable assistance for English learners who are specifically focused on developing their academic writing skills for educational purposes (Erkan, 2022; Nurmayanti & Suryadi, 2023; Tran & Nguyen, 2021). Numerous studies have provided evidence to support the notion that the incorporation of AI discussion chatbots in educational settings has yielded notable benefits in terms of enhancing students' English communication proficiency. This positive outcome can be attributed to the chatbots' ability to provide more precise language feedback, as well as their capacity to create additional opportunities for interactive engagement (Junaidi, 2020; N.-Y. Kim, Cha, & Kim, 2019).

2.2. Training Programming Skills with AI Assistance

There has been a research stream that investigates the possible benefits of using AI systems in schools (L. Chen et al., 2020). One of the main perks is that learning goes faster and is more accurate. AI systems can quickly look through huge amounts of data and find complex patterns and links that students might find hard to see due to the lack of experiences and hints. AI systems can also help make personalized learning paths (X. Chen, Cheng, Zou, Zhong, & Xie, 2023; Hsu, Huang, Hwang, & Chen, 2023). More specifically, by looking at how and why students learn, AI systems can give each student personalized help and advice that fits their specific needs. This makes it easier for them to reach their educational goals. AI systems can also make online learning easier, which means that students who live in remote places can get the same access to education as other students (Akgun & Greenhow, 2021; Chichekian & Benteux, 2022). This encourages justice and inclusion by letting everyone get a chance of better education.

Concurrently, there are continuous deliberations concerning the obstacles that necessitate resolution for the effective implementation AI applications and services within the context of education. One of the prominent topics of concern pertains to the challenges that arise when AI incorporates preexisting biases or discriminatory inclinations. Moreover, a contentious issue arises over the extent to which proficiently trained AI can deliver educational content that is both truthful and reliable. Critics contend that the progress in empirical evidence and the formulation of experiential theories pertaining to the implementation of AI in the field of education lags behind the rapid breakthroughs witnessed in data science areas, shown by ChatGPT. Despite being in the preliminary phase, research conducted by Akgun and Greenhow (Akgun & Greenhow, 2021) on elementary and secondary school students, as well as studies by Okonkwo and Adelbijola (Okonkwo & Ade-Ibijola, 2021) examining the ethical use of AI chatbot systems in higher education, all call for a comprehensive examination of the potential challenges that may arise with the active integration of AI as an educational tool.

Zhang and Aslan (Zhang & Aslan, 2021) conducted an extensive investigation on the application of AI in education and its potential advantages. Their study highlighted several benefits, such as increased student engagement and motivation, enhanced testing procedures with immediate feedback, as well as improved efficiency and cost-effectiveness of the educational process. Similarly, Pedro et al. (Pedro, Subosa, Rivas, & Valverde, 2019) discussed the merits of integrating AI into education in their scholarly article. The authors concurred that leveraging AI technology in educational settings could automate mundane tasks, allocate more time to challenging cases, and equip individuals with new skills necessary for effectively collaborating with AI systems. Pedro et al. (Pedro et al., 2019) discussed the challenges and policy implications associated with the implementation of AI in educational settings and the preparation of students for real-world situations. The authors identified several key issues that require attention, including ensuring equitable and inclusive access for all individuals. According to Pedro et al. (Pedro et al., 2019), concerns regarding AI in education encompass unequal access to educational systems resulting from machine learning algorithms, inadequate guidance provided to specific student groups, excessive retention of personal data, liability concerns, and the potential impact of AI automation on teachers' roles. Safeguarding personally identifiable information and privacy settings poses a significant challenge, particularly when dealing with young students who may not possess the capacity to provide informed consent.

Programming with the help of artificial intelligence provides a certain level of freedom in ethical concerns regarding the content, as AI can serve as a limited guide by combining publicly available algorithms, code snippets, documentation, and development knowledge to provide explanations or showcase exemplary or sample code. While using AI in coding education may not completely solve or eliminate all issues related to the use of AI in educational settings, considering the practical environment and conditions of current generative AI, it is believed that it can provide substantial assistance in university students' education while also offering a rich information environment for information system researchers.

3. Methodology

3.1. Tasks

The field of programming is characterized by its complexity, requiring individuals to acquire a comprehensive understanding of various concepts and structural regulations to effectively study and implement its principles. Xie et al. (Xie et al., 2019) devised an instructional theory that was specifically designed for enhancing the effectiveness of teaching programming skills in introductory courses. This theory suggests that students should learn specific skills in small steps so that they don't get too overwhelmed. It can be challenging for students to apply a concept directly to programming, even if they have learned it. Teaching programming solely through lecture-based approaches may not be effective as it expects students to grasp unfamiliar concepts all at once (Shaw, 2017).

Xie et al. (Xie et al., 2019) proposed a method of gradually combining small steps, which involve skills such as reading semantics, writing semantics, reading models, and writing models. Table 1 shows the characteristics of each step.

Step	Learning goal
Reading Semantics	Getting the capacity to precisely track code and anticipate the impact of syntax on program behavior
Writing Semantics	Acquiring abilities of converting clear and precise natural language descriptions of language constructs into syntax that is capable of being compiled and executed as intended
Reading Templates	Getting the ability to recognize and apply reusable abstractions of programming knowledge, referred to as templates, and aligning them with a certain aim
Writing Templates	Acquiring the capacity to discern a suitable structure of potential templates and build a new template by acquired expertise and knowledge

Table 1. An instruction model for programming learners suggested by Xie et al. (Xie et al., 2019)

To assess the academic performance of the students, they were instructed to complete 10 quizzes, each corresponding to a specific learning module. These quizzes were evaluated by checking their responses. The participants were assigned the task of acquiring proficiency in using the *R* programming package known as *dplyr*, which

is specifically designed for data manipulation. This learning process spanned a period of two weeks. After the completion of each module, quizzes were administered to record the grades achieved by the students.

The initial step of **Reading Semantics** aims to provide a theoretical understanding of fundamental concepts. Students learn about data and key functions, utilizing the data package `nycflights13` which contains information on flight departures and arrivals in New York airports. Additionally, the characteristics and significance of *dplyr* functions are explained, although the direct usage of these functions is not demonstrated. Instead, emphasis is placed on illustrating how inputs and outputs are affected by the functionalities performed by these functions with text descriptions. In the phase of **Writing Semantics**, the primary objective is to facilitate the conversion of lucid and meticulous natural language explanations of language constructs into a syntactic form that can be compiled and executed as intended. In this process, students engage in the analysis of input and output tables presented in Excel as illustrative instances. They deliberate upon the necessary steps involved in converting input data into desired output, while also exploring the means to achieve this objective through the utilization of *dplyr*'s function names. **Reading Templates** consist of source code and comments explaining how they work. They provide brief glimpses into the functioning principles of the key functions within *dplyr* and their resulting outcomes by showcasing various examples. Additionally, they also explain basic arithmetic functions included in R and provide descriptions of *data.frame* objects. Finally, in the **Writing Templates**, students are required to perform data manipulation tasks using the `nycflights13` dataset and utilize RStudio provided by *posit.cloud* to write RScript files. Figure XXX provides a brief illustration of the learning process for combining `select()`, `filter()`, and `mutate()` functions in R.

<<Table 2 HERE>>

Step	Tasks
Reading Semantics	Learning about <code>nycflights13</code> data sets and functions including <code>select()</code> , <code>filter()</code> , <code>arrange()</code> , <code>group_by()</code> , <code>summarise()</code> , <code>mutate()</code> , <code>ungroup()</code> , <code>inner_join()</code> , <code>left_join()</code> , and <code>n()</code> but not limited to
Writing Semantics	Developing logical flows to construct new data sets to address data analysis requirements using diagrams supplied by PowerPoint and text descriptions written by students
Reading Templates	Learning about examples that demonstrate how R <i>base</i> and <i>dplyr</i> functions are working by presenting input and output results
Writing Templates	Developing actual R scripts to address data analysis requirements based on what students have learned

Table 2. Example steps of learning dplyr functions in class by Xie et al. (2019)'s approach

3.2. Treatment

During the process of learning programming and coding skills, some students receive assistance from generative LLM (large language model)-based artificial intelligence, while others do not. The students are divided into three groups: one group receives assistance from generative artificial intelligence, another group does not receive such

assistance, and the third group receives help from the Stack Overflow website. The LLM artificial intelligence used in this study is ChatGPT, a service provided by OpenAI. This service is available both as a paid and free version, and all participants utilized the free version, specifically ChatGPT 3.5. The comparison groups consisted of participants who received support from ChatGPT and those who did not, with the group obtaining information from Stack Overflow serving as the control group. According to the survey, participants reported having a nearly identical experience with ChatGPT. ChatGPT was officially introduced in the late of 2022 and is based on artificial intelligence that can mimic human language expression to engage in conversations at a human-like level. Additionally, it can provide explanations of the program's structure and find necessary functions and example code based on user queries. However, it should be noted that the reliability of the results is not perfect, as ChatGPT may generate inaccurate answers or produce non-executable code.

Table 3 depicts a collection of sample prompts related to the R function `lm()`. The course did not provide students with materials on prompt engineering techniques. Instead, students were informed about the potential risks associated with hallucination issues in ChatGPT. As of early 2023, when the data was collected, the performance of ChatGPT remains inconclusive. The code may encounter occasional failures during execution, and there is no explicit warning in the output regarding this risk.

Human (prompt)	ChatGPT (response)
"I am using R. Explain me about <code>lm()</code> ."	<i>In R, the <code>lm()</code> function is used to fit linear models. It stands for "linear model," and it is a fundamental tool for performing linear regression analysis. Linear regression is a statistical method that models the relationship between a dependent variable and one or more independent variables by fitting a linear equation to the observed data.</i> ... <i>The basic syntax for using <code>lm()</code> is as follows:</i> ... <i>For example, if you have a dataset <code>my_data</code> with variables <code>y</code>, <code>x1</code>, and <code>x2</code>, and you want to fit a linear model relating <code>y</code> to <code>x1</code> and <code>x2</code>, you would use:</i> <code>my_model <- lm(y ~ x1 + x2, data = my_data)</code> <i>The <code>~</code> symbol is commonly used in R formula notation to specify the relationship between the dependent and independent variables in a model.</i> ...
"What is the tilt (<code>~</code>) in your sample codes?"	... <i>The <code>~</code> symbol is commonly used in R formula notation to specify the relationship between the dependent and independent variables in a model.</i> ...

Table 3. Example of prompts to learn functions for OLS regression

With ChatGPT, students can leverage AI to enhance their understanding of concepts and receive explanations with various examples. It's akin to having a personal tutor guiding them through Xe et al.'s (Xie et al., 2019) four-step learning process. Students will be able to clear their doubts more quickly, leading to an expected improvement in learning

efficiency and performance. Unlike reading a book or consulting online resources, ChatGPT offers an interactive experience that can enhance student engagement in learning by making it more interesting and familiar.

3.3. Participants

The experimental participants consisted of first- and second-year university students, specifically business administration majors at a university in Seoul, South Korea. They were divided into two groups based on their class attendance period: one group attended classes from September to December 2022, while the other group studied from March to June 2023. The distribution of the participant groups is presented in Table 4.

Category	Subcategory	Code Assistant	N	Previous GPA	Programming Experience
Group A	Group A-1	-	110	3.7 (0.2)	0.18 (0.39)
	Group A-2	Stack Overflow	109	3.6 (0.1)	0.21 (0.41)
Group B		ChatGPT	124	3.4 (0.3)	0.71 (0.38)

A perfect GPA is 4.5. The displayed score represents the average GPA for the courses taken in the previous semester. Programming Experience is calculated as the average of scores for items coded as 1 for having experience with any type of programming, and 0 otherwise. Standard deviation values are indicated in parentheses.

Table 4. Experiment settings: code assistant impact on Groups A and B

The course being offered in this study was titled "Business Data Analytics," which was an elective course for students pursuing their respective majors. The class consists of multiple groups, each comprising a similar number of students (N = 110, 109 and 124 respectively). The instructional format included online lectures and offline labs. It is worth noting that the average grades across the different student groups were relatively consistent. Additionally, based on survey responses regarding programming experience, it was found that less than 50% of the students reported having no prior programming background.

It is important to highlight that the data collected from Group A during October 2022 was not pertinent to ChatGPT due to its temporal context. Conversely, the data obtained from Group B in April 2023 holds relevance to ChatGPT as it was collected when ChatGPT had gained widespread recognition. Moreover, during this period, students had the opportunity to leverage ChatGPT as one of the lectures specifically addressed its usage within the lecture.

3.4. Quiz items to track performance

To assess student learning for each stage, we originally administered five multiple-choice quizzes in which students were required to choose one option out of two. For example, a question may ask a student to choose an appropriate function in *dplyr* that is used to select columns in a data table, giving them the options of `select()` and `subset()`. In this study, we adopted Item Response Theory (IRT) to develop valid test items to evaluate the level of learning for each step objectively (Edelen & Reeve, 2007). When all assumptions of Item Response Theory (IRT) were met, the data was fitted with an IRT model to estimate discrimination (α) and threshold (β) parameters. This estimation was performed using the expectation-maximization algorithm implemented in the 'mirt' R package (Chalmers, 2012).

The α parameter is utilized to measure the item's ability to differentiate between participants with varying levels of performance (θ), while the β parameters represent other characteristics. Figure 2 shows item infit and outfit statistics graphically. Items with values within 0.5 and 1.5 are considered to be productive for measurement. The V4 items in the Writing Semantics step were excluded from the comparison because they had a poor fit value (output = 0.180, z.output = -3.381, infit = 0.385, z.infit = -6.042) as shown in Figure 2.

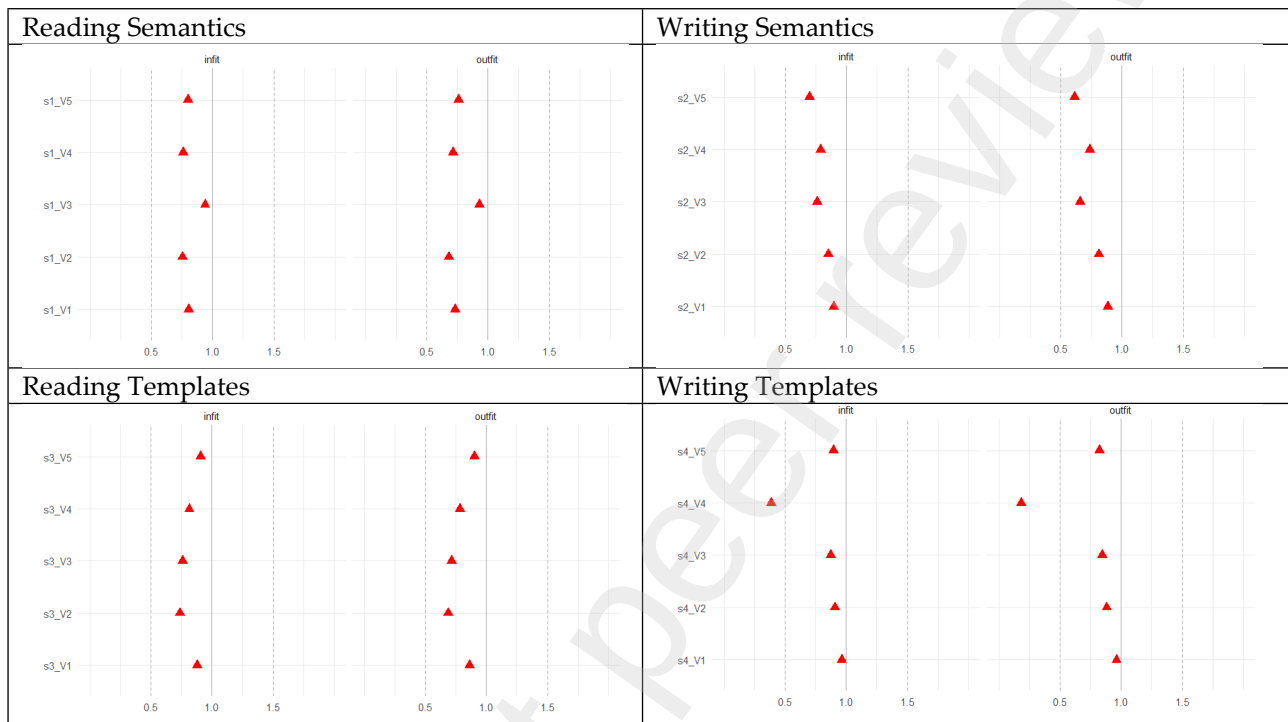


Figure 2. Infit and outfit statistics of quiz items to measure learning performance

Stage	Item ID	S^2_X (p-value)	Stage	Item ID	S^2_X (p-value)
Reading Semantics	S1_V1	3.714 (0.156)	Writing Semantics	S2_V1	0.734 (0.693)
	S1_V2	1.457 (0.483)		S2_V2	0.168 (0.92)
	S1_V3	2.673 (0.263)		S2_V3	2.091 (0.351)
	S1_V4	4.199 (0.123)		S2_V4	1.901 (0.386)
	S1_V5	0.248 (0.883)		S2_V5	0.065 (0.968)
Reading Templates	S3_V1	0.996 (0.608)	Writing Templates	S4_V1	1.347 (0.51)
	S3_V2	1.193 (0.551)		S4_V2	0.788 (0.674)
	S3_V3	0.77 (0.68)		S4_V3	3.099 (0.212)
	S3_V4	0.241 (0.886)		S4_V4	0.849 (0.654)
	S3_V5	0.987 (0.611)		S4_V5	2.036 (0.361)

Table 5. Results of item fit computation by stage

4. Results

4.1. Performance

In this section, we will introduce the results of the experiment and summarize their implications for discussion. A regression model was analyzed using the OLS method to examine the differences between three groups: Group (i.e., C)

that only referred to course materials to understand the usage of Stack Overflow, Group (i.e., B) that received education on how to utilize Stack Overflow, and Group (i.e., A) that utilized ChatGPT-3.5. Table 6 presents the treatment effects of Group A and Group B relative to Group C as the reference group.

First, obtaining help information about the *dplyr* package in R from Stack Overflow was found to be useful in learning ($\beta = 0.486$, p-value < 0.05). Students were observed to utilize search engines like Google or Microsoft Bing, as well as the built-in search feature of Stack Overflow, to find relevant information. As the classes were conducted in an interactive manner, the process of students' assignment completion was closely observed individually. Through discussions in a team, students gained an understanding of the learning tasks and then looked for examples or acquired ideas for resolving errors encountered during code execution on Stack Overflow.

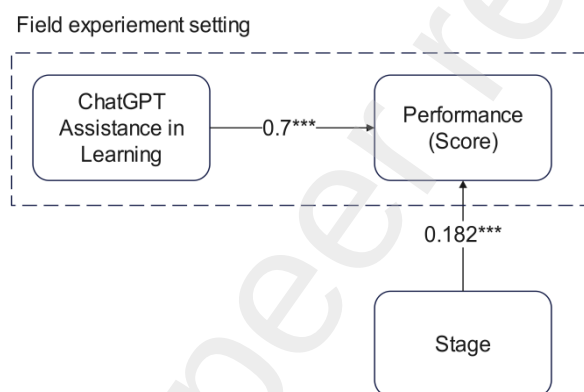


Figure 3. Effect of adopting ChatGPT as a learning assistant in programming class

On the other hand, the use of ChatGPT appears to be more beneficial in acquiring coding skills compared to Stack Overflow ($\beta = 0.7$, p-value < 0.001). The average score when using ChatGPT was 60.04%, which is higher than the baseline of 46.04% and even higher than Stack Overflow's 52.6%. Overall, generative artificial intelligence can greatly assist students in learning how to use complex data processing packages like *dplyr* in R.

During the Reading Semantics stage, the scores of the C and B groups did not show a statistically significant difference (t-value = 1.38, p-value = 0.168). This lack of significant difference was also observed in the Writing Semantics stage (t-value = 0.76, p-value = 0.448). However, it is worth noting that there was a significant average difference in scores between these two groups, with a mean difference of 13.42 (t-value = 13.14, p-value < 0.001) for Reading Templates and 3.36 (t-value = 3.9, p-value < 0.001) for Writing Templates.

When comparing the mean values, a notable disparity was observed between the group receiving assistance from ChatGPT and the other two groups. Group C, which utilized ChatGPT throughout all stages, demonstrated superior performance across all metrics. This discrepancy is visually depicted in Figure 4. Following Xi et al.'s (Xie et al., 2019) instructional protocol, evaluation scores were conducted at the conclusion of each stage in the coding education program. Group C, who solely received lectures for coding education, experienced a gradual decline in scores. While they achieved the highest scores in the Reading Semantics stage, which emphasizes comprehension of concepts, their

lowest scores were observed in the Writing Templates stage, where independent coding was required. Conversely, Group A, utilizing ChatGPT, exhibited contrasting patterns. They consistently attained higher scores starting from the conceptual understanding stage, with the most significant score difference observed during the self-coding phase using Writing Templates. Since they did not directly receive assistance from ChatGPT during quizzes, this disparity cannot be attributed to the presence or absence of information technologies. The ability to seek assistance from ChatGPT during exam preparation proved to be crucial. In essence, progressing through the stages with AI support for coding learning consistently yielded superior outcomes compared to relying solely on lectures.

OLS regression results:			
DV: Sum of scores	Estimated coefficient (standard error)		T-value
ChatGPT use	0.7 (0.087)		8.011***
Stack Overflow use	0.486 (0.285)		2.386*
Step	0.182 (0.039)		4.647***
ANOVA results:			
	Sum of squares	Mean of squares	F value
ChatGPT use	122.5	122.5	64.17***
Stack Overflow use	41.2	41.22	21.59***
Residuals	1903.2	1.91	
Comparisons of mean scores by stage			
Stage	Base	Stack Overflow	ChatGPT
Reading Semantics	50.08 %	51.14 %	57.92 %
Writing Semantics	40.48 %	41.72 %	59.04 %
Reading Templates	42.40 %	55.82 %	58.72 %
Writing Templates	39.04 %	42.40 %	64.48 %

Table 6. Learning performance analysis results

The comparison between Stack Overflow and ChatGPT reveals intriguing findings. Encouraging students to utilize Stack Overflow for studying proved to be more beneficial in comprehending abstract concepts rather than practical coding. This disparity can be attributed to inherent differences between the platforms. Furthermore, interview results presented in subsequent sections highlighted that students encountered difficulties when attempting to implement code obtained from Stack Overflow, indicating that a lack of coding experience adversely affects academic performance. It is worth noting that the students included in the study were not computer science majors but instead possessed backgrounds in business studies, which rendered them unfamiliar with the technical jargon commonly used by professional developers on Stack Overflow. In essence, while studying via Stack Overflow may contribute to achieving high scores in theoretical examinations, engaging in practical discussions with ChatGPT proves more advantageous for attaining true mastery of coding skills.

4.2. Results of analytic hierarchical processing

Our research findings indicate that the utilization of ChatGPT positively impacted academic performance across all stages of learning. To gain a deeper understanding of this phenomenon, we conducted supplementary interviews

with a sample of 24 students. These interviews aimed to investigate the differences between using ChatGPT as a collaborative tool, combined with the knowledge repository of Stack Overflow, versus relying solely on traditional classroom-based learning methods. In this study, the evaluation of learning methods has involved considering factors such as ease of use, usefulness, and satisfaction, drawing from concepts in the technology acceptance literature (Phan & Daim, 2011; Saadi, Ahmad, & Hussain, 2017; Venkatesh, Morris, Davis, & Davis, 2003). To comprehensively evaluate the process of selecting learning methods in our study, we administered a survey alongside the interviews. This allowed us to gather detailed information on alternative judgments and subsequently analyze this data using the Analytic Hierarchy Process (AHP) method.

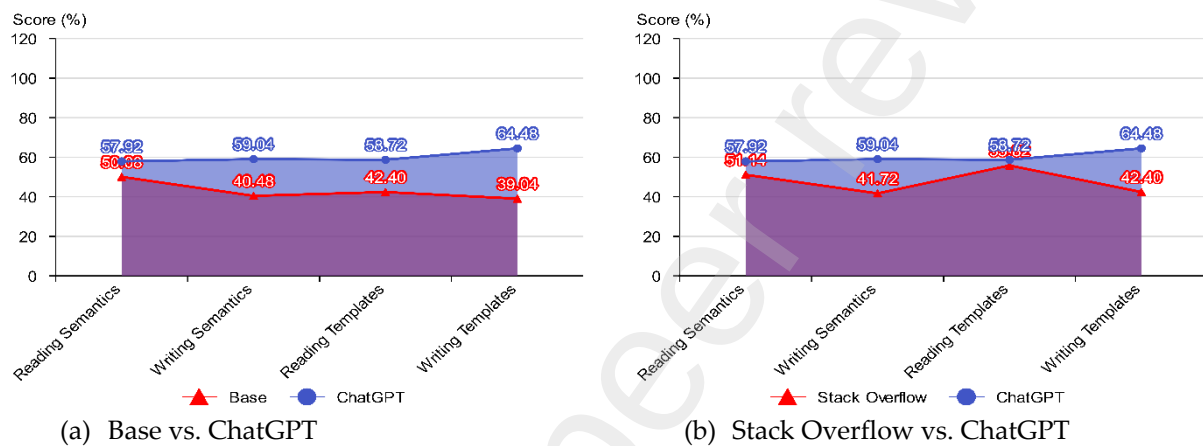


Figure 4. Comparative mean values demonstrating ChatGTP's impact

AHP facilitates the partitioning of a decision problem into a hierarchical structure consisting of numerous qualities or criteria. This hierarchical structure is designed such that the higher-level parts encompass more specific elements, hence increasing the amount of detail and granularity (Saaty, 1990, 2008; Zahedi, 1986). AHP has demonstrated its effectiveness in addressing a wide range of intricate decision-making challenges (Zahedi, 1986). Numerous previous studies have employed AHP for the goal of ranking various entities. Colace et al. (Colace, De Santo, & Pietrosanto, 2006) applied the Analytic Hierarchy Process (AHP) method while studying an e-learning platform. Chen Hsieh & Do (J.-F. Chen, Hsieh, & Do, 2015) introduce a novel framework for assessing teaching performance in higher education. This framework integrates the fuzzy Analytic Hierarchy Process (AHP) and fuzzy comprehensive evaluation methods to establish a teaching performance index system.

The Analytic Hierarchy Process (AHP) is a versatile methodology that can be applied to address a range of practical problems. Examples include the selection of suitable building sites, the development of evaluation criteria for performance appraisals, the formulation of public policies, and the decision-making process for launching new products in the market. AHP employs a ratio scale to measure relative importance or preferences, enabling the derivation of quantitative results. The execution of AHP involves several steps. Firstly, the problem's goal is defined. Subsequently, a hierarchical structure is constructed, encompassing criteria and alternatives for evaluation. Pairwise comparison matrices are then created using reciprocal ratios. These matrices serve as the basis for calculating relative importance

and assessing coherence and consistency through the computation of a consistency index (CI). If the CI falls below an acceptable threshold, further discussions with respondents are necessary to address any inconsistencies. Finally, eigenvalues derived from these calculations are synthesized to determine the relative importance among alternative options.

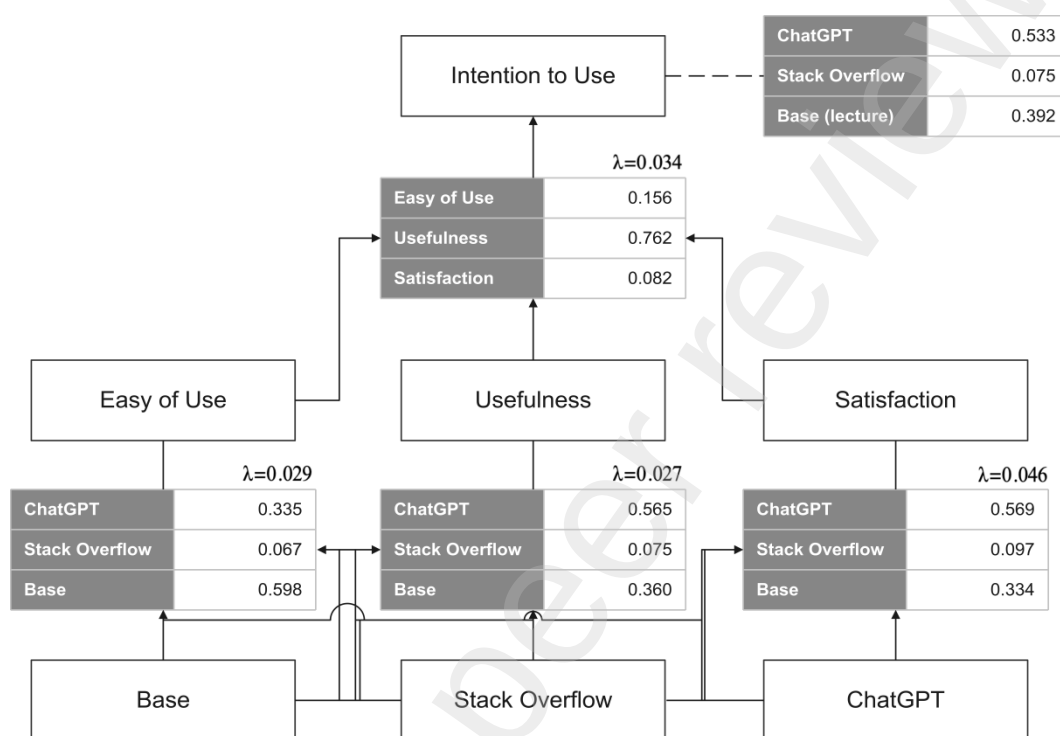


Figure 5. Results of analytical hierarchical processing

The initial inquiry pertains to the extent to which ease of use aids learners in their learning process. We conducted interviews in which participants were requested to compare and assess the relative significance of traditional lectures (Base), support through Stack Overflow, and utilization of ChatGPT on a scale ranging from 1 to 9, with regards to ease of use. In the analysis, the comparative outcomes were expressed as standardized eigenvalues that range from 0 to 1. The consistency index (CI) was computed based on the eigenvalues provided by each respondent, and if it was below 0.1, the responses were considered consistent. To calculate the CI promptly as participants provided their answers, we developed an R-shiny app and re-evaluated any areas where response consistency was compromised. Additionally, we inquired about the reasons underlying these assessments. From the perspective of ease of use, lectures were perceived as having the highest importance in facilitating learning (0.598), while Stack Overflow was regarded as presenting the greatest challenge (0.067).

In the evaluation of the usefulness aspect, a similar approach was applied as before, where respondents were asked to rate the relative importance on a scale of 1 to 9 and provide reasons for their ratings. The average lambda values, which indicate response consistency, were below 0.1, showing consistent responses. The results revealed that ChatGPT (0.565), Base (0.360), and Stack Overflow (0.075) were ranked in that order.

In the evaluation of the relative importance of each learning alternative in terms of satisfaction, similar to usefulness, ChatGPT received an importance score of 0.569, followed by learning through lectures with a score of 0.334, and Stack Overflow with a score of 0.097. We asked the interviewees to define satisfaction in their own words, and the satisfaction with coding support processes was found to be related to immediacy, comprehensibility, and reproducibility. Specifically, the ability to quickly utilize the outcomes obtained through learning was frequently mentioned as an important motivating factor for satisfaction with their choice.

According to the provided criteria, an assessment was conducted to determine the relative importance of alternatives. Subsequently, a pairwise comparison was made to determine the relative importance of criteria for intention to use as the target. The findings indicate that the usefulness criterion held the highest level of importance (0.762), followed by ease of use (0.156), with satisfaction ranking lower (0.082). It is worth noting that the interviewees were novice users of R learning coding in university classes, suggesting a notable inclination towards methods that can impact learning performance evaluation.

Based on the relative importance of criteria, the expected contributions of alternatives to the goal were evaluated. The results showed that ChatGPT had the highest score of 0.533, followed by traditional lecture methods (0.392), and finally Stack Overflow with a score of 0.075.

In the previous session, when measuring performance by learning stage, both Base and Stack Overflow achieved similar scores. However, Stack Overflow showed relatively disappointing results in the Writing Semantics and Writing Templates stages. The results from AHP shed light on this. While Stack Overflow, which is centered around Q&A, proves helpful in providing tips or checking solution options for specific topics, it seems to fall short in providing comprehensive information needed when learners attempt to code themselves. On the other hand, ChatGPT demonstrated more consistent results in terms of usefulness in learning due to its ability to respond to learners' questions with greater flexibility. Detailed testimonies and explanations related to this will be provided in the following sections.

4.3. Reasons of relative preference to ChatGPT in coding learning

The interview results with 24 students, conducted to gather data for AHP analysis, consistently highlighted the enjoyment and companion-like feeling of using ChatGPT, as well as its kindness, as significant advantages. It was noted that ChatGPT offers a noticeably different experience compared to traditional means of accessing information or simply relying on web search engine results. The notion of considering ChatGPT as a close peer to overcome the challenges of learning together is particularly intriguing.

As a business analytics major, not really coming from a computer science background, I've gotta say, getting good at programming is pretty tough. It's like, I know it's important, but it's definitely a challenge to pick up all these coding skills when you're not in a hardcore tech major, you know? Talking to my professor definitely helps, but they're usually so swamped, it's hard to get a lot of time with them. That's where ChatGPT comes in super handy. It's always there to give me detailed answers. Even

when I'm totally lost on some topics, I just keep asking questions, and studying feels less overwhelming. It's like having this 24/7 helpdesk that makes everything clearer for me.

It can be challenging to determine what to study when handling data and understanding the procedures involved, especially when it's a topic that students are not familiar with. Initially, understanding what ChatGPT is saying and knowing what questions to ask can be difficult, and searching on Stack Overflow may also pose challenges in terms of interpreting the results and comprehending complex information. However, having lecture slides in PowerPoint format with explanations from the professor can be helpful due to their clear examples.

Xie et al. (2019) recommended a structured approach to programming learning, segmenting it into Reading Semantics, Writing Semantics, Reading Templates, and Writing Templates. They advocated for a gradual progression through these stages. In teaching coding with R for data manipulation, we adopted this learning sequence. We also facilitated access to Stack Overflow and ChatGPT for learner support. Our students appreciated traditional lecture-based methods for Reading Semantics and Writing Semantics. However, they found generative AI support immensely beneficial for hands-on coding in the Writing Templates phase.

Whenever I need examples, ChatGPT is right on it. And the best part? If I need something tweaked, it doesn't just tell me – it shows me the code and the results. I don't have to test each example myself. It's been awesome for understanding how the code really works, especially with different functions. Like, when I was messing around with `str_match_all()` in `stringr` and got stuck with regular expressions, ChatGPT came through with spot-on examples and easy-to-get explanations. Seriously, it's made studying a whole lot easier for me.

The stuff my professor shows in class? Super clear and really helpful. But, when I try coding on my own, it's mistake city. It's pretty embarrassing, honestly. Asking others what I'm doing wrong is tough, and even with some Stack Overflow help, I end up Googling like crazy to figure out my errors. I didn't want to bug my friends with non-stop questions, and finding help late at night? Almost impossible. Emailing professors is tricky too, 'cause sometimes I don't even know what to ask. But ChatGPT? It's a game-changer. It gets me, even when my questions are kind of vague. I drop in some code or a half-baked explanation, and it just rolls with it, giving me exactly the kind of conversation I need.

A common impression we gathered during the interviews is that students were fascinated by the utility provided by ChatGPT. They also emphasized the ongoing need for lectures and communication with professors. While Stack Overflow offers many useful tips, having interactive features that provide more examples and explanations for each problem proved to be more helpful. Having a companion who engages in coding alongside interactive explanations, rather than just searching and reading from a knowledge repository, was greatly appreciated.

5. Concluding Remarks

The study conducted by Wang et al. (2023) presents a comprehensive investigation into the impact of artificial intelligence on international student education. The researchers thoroughly explore various applications of AI in this context and highlight significant considerations such as privacy, cultural differences, language skills, and ethical implications. Similar to their findings, our research also revealed that the capabilities of generative AI provide significant assistance to learners. However, unlike Wang et al. (2023), we compared long-term learning processes and quantitatively assessed the areas in which learners benefited based on decision models, while also striving to gain a more detailed understanding through interviews.

The impact of utilizing an AI chatbot in the learning process was observed to be significant. This was evident through its ability to provide a wide range of coding examples, thereby enhancing students' understanding. Additionally, the interactive nature of the chatbot encouraged students to ask questions actively, leading to increased interest and enjoyment in the learning experience. The integration of artificial intelligence in educational settings goes beyond this specific instance and includes personalized learning approaches, intelligent tutoring systems, and improved monitoring of the learning process and interactions between learners and teachers (Kamalov et al., 2023). Our empirical research findings demonstrate the considerable value of generative artificial intelligence based on LLM for educational or training purposes. We believe that it holds promising potential in meeting the demand for high-quality higher education amidst anticipated population decline.

It is crucial to recognize the limitations of this study. The data analysis findings were derived from a specific context, namely, the teaching of R for data analysis to business school students. Given the homogeneity of these students in terms of skills and learning backgrounds, it can be argued that the internal validity and reliability of the results are relatively robust. However, caution should be exercised when attempting to generalize these findings to other programming learning tasks or students with diverse backgrounds. Further evidence is required to support such generalizations.

This study did not explore the potential combination of more traditional learning methods with generative AI. Methods such as flipped learning, collaborative learning, and project-based learning could further enhance the possibilities of generative AI. However, as we experimented with the potential of generative AI as a supplementary learning tool in a traditional lecture-based setting, we did not provide clues about such outcomes. Further research is needed to address these limitations.

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Generative AI, Chatbots, and Programming Skills: Examining Code Suggestion Features in
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